

Wang Shaohui

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Zhengzhou University • M.S. Sports Science • 2023.09-2026.06

Profile: A data-driven testing engineer bridging human physiology, data science, and standardized testing — from sensor data collection to algorithm validation

Education

EDUCATION

Zhengzhou University - Master of Sports Science (Sports Training)

2023.09 - 2026.06

GPA: 3.89/4.3 • Research Focus: Physical fitness promotion, sports performance enhancement, sports data analysis

- Core courses: Theory & Methods of Sports Training (96), Physical Education Pedagogy (93), Professional English (94)
- Systematic training in sports data collection and analysis methodologies; familiar with sensor-based physiological signal (HR, SpO2, accelerometer) acquisition and processing workflows

Zhengzhou University - Bachelor of Physical Education

2019.09 - 2023.06

GPA: 2.99/4.0 • Core courses: Sports Training (90), Motor Learning & Control (90), Basketball (93)

Work Experience

EXPERIENCE

Suunto - Algorithm Test Engineer

2026.07 - Present

- Test Case Design & Validation:** Designed and validated test cases for core wearable-device algorithms (heart rate, SpO2, sleep, sport modes, barometer); built standardized end-to-end testing workflows covering the full algorithm pipeline.
- Quantitative Analysis:** Conducted quantitative analysis of algorithm outputs using real-world exercise data; identified and localized defects in HR detection and sleep staging algorithms; collaborated with the algorithm team on fix verification.
- Acceptance Criteria:** Partnered with algorithm and product teams to define acceptance criteria and core evaluation metrics (accuracy, recall, etc.) for each algorithm module.
- Evaluation Framework:** Built an algorithm performance evaluation framework; benchmarked algorithm versions on key metrics and produced evaluation reports to support product iteration.
- AI-Driven Workflow:** Introduced AI agents into the testing workflow for automated test-case generation, defect report consolidation, and process orchestration — markedly improving testing efficiency and demonstrating AI deployment in real business scenarios.

Publications

PUBLICATIONS

Effects of three strength training methods on lower extremity strength, jump and sprint performance: A network meta-analysis

Frontiers in Physiology JCR Q2 — Built an effect network across three lower-extremity strength training methods, providing quantitative support for sports performance enhancement.

Combined training outperforms single-form in improving lower extremity reactive strength index: A network meta-analysis

Journal of Science in Sport and Exercise JCR Q3 — First systematic comparison of combined training vs. single-form plyometric training effects on RSI.

- Conferences: National Sports Science Congress, National Competitive Sports Science Congress, Asia-Pacific Sports & Exercise Science Congress.

Research Projects

PROJECTS

- Ministry of Education Humanities & Social Sciences Project**
Research on college students' physical health crisis and mitigation strategies under the Healthy China initiative
- Henan Provincial Philosophy & Social Sciences Planning Project**
Network-based intervention and RE-AIM evaluation of college students' physical health behaviors in Henan Province
- Ministry of Education Industry-Academia Collaborative Education Project**
Strategies for improving students' physical health under sports-education integration
- University-Level Projects**
Seven projects including Zhengzhou University first-class courses and education reform initiatives

Professional Skills

SKILLS

Human Factors Data Collection

- HR / SpO2 / accelerometer / barometer multi-sensor signal acquisition
- Human-subject testing protocol design and execution
- Wearable-device algorithm full-pipeline validation

Data Analysis & Statistics

- Python (scikit-learn) statistical modeling & ML
- R (tidyverse / meta) network meta-analysis
- SPSS, Stata — comprehensive data analysis
- Git version control and collaboration

Algorithm Evaluation & AI

- Accuracy / Recall / F1 / κ-coefficient evaluation framework
- Bland-Altman agreement analysis
- AI agents applied to testing workflows
- End-to-end capability from theory to engineering

Additional Experience

PRACTICE

- Part-time PE Teacher:** Taught physical education during undergraduate studies, developing strong communication skills.
- Sports Coach:** Provided sports training guidance, building management and coordination abilities.
- Research Paper Tutoring:** Tutored peers in thesis data analysis and writing, demonstrating cross-domain knowledge integration and communication.

Human Factors Experiment & Data Analysis Engineer